Assignment 4

Forms and Validation · HTML + Bootstrap

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GitHub: https://github.com/tamerlanaitpayev1-netizen/web4

This assignment implements three forms: a train ticket layout based on the reference, a hackathon registration form with native constraints, and a corrected cafe reservation form. Bootstrap and Montserrat are included locally. The pages use no JavaScript and no novalidate attribute.

1. Train ticket booking

The responsive layout uses row, col-12, col-md-6 and col-md-4. Form components include form-control, form-select, btn-check, form-check, form-switch and input-group. Custom CSS provides the required colors, rounded corners, shadows and spacing. The task CSS uses no fixed widths or absolute positioning.

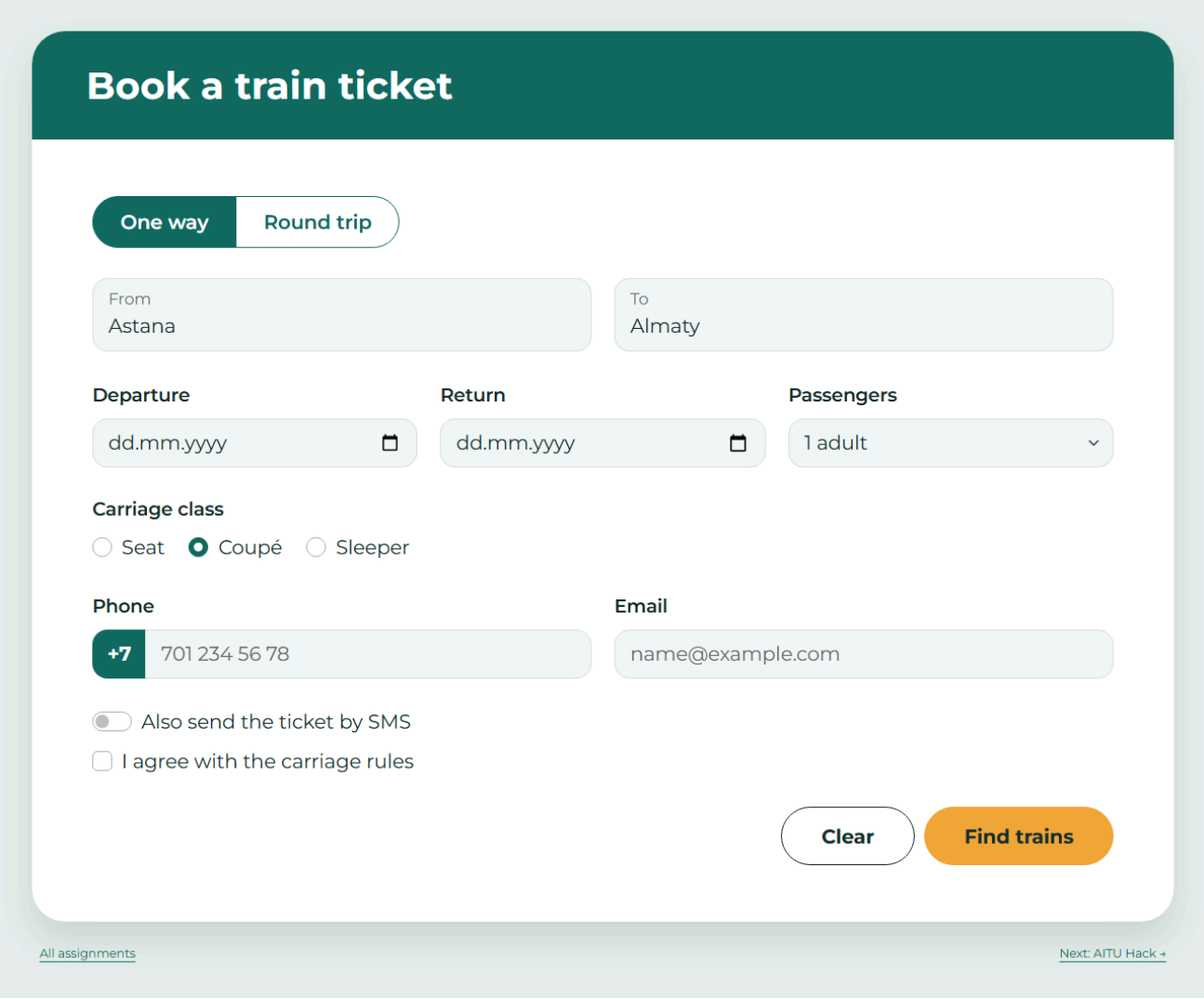


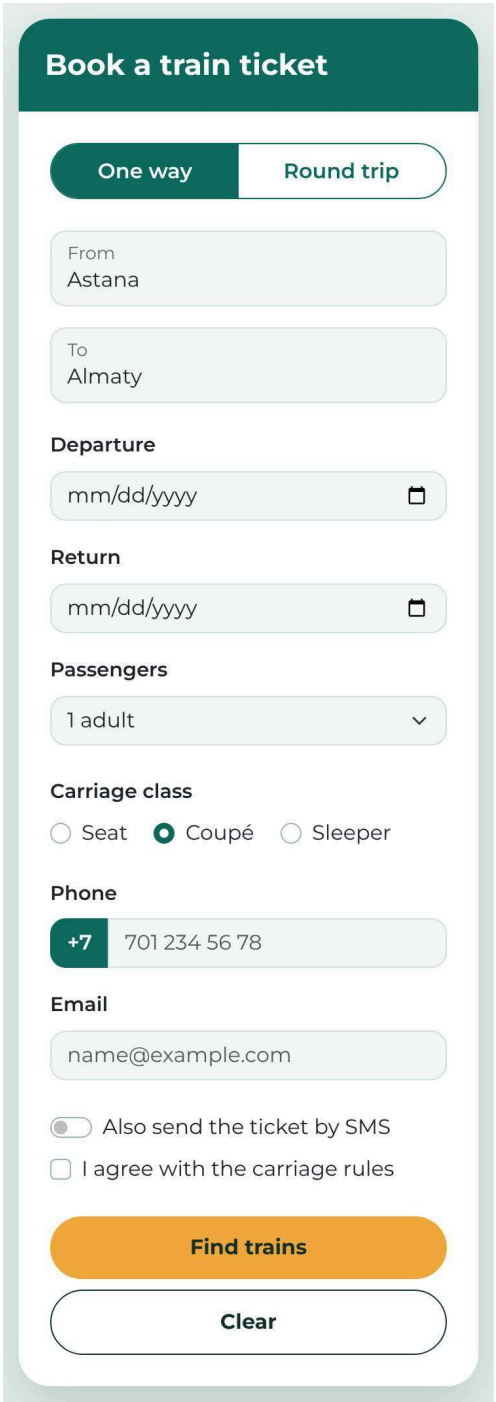
Figure 1. Task 1 at a viewport width of 1280 px. This populated example selects One way, Coupé and 1 adult, with Astana / Almaty entered; the initial form is empty.

The route, departure date, carriage class, phone, email and agreement are required. The passenger list offers 1–6 adults. Return is optional. Labels use matching for/id values. The native reset button clears entries and selections because the initial form contains no default value/checked attributes; the fixed +7 telephone prefix remains.

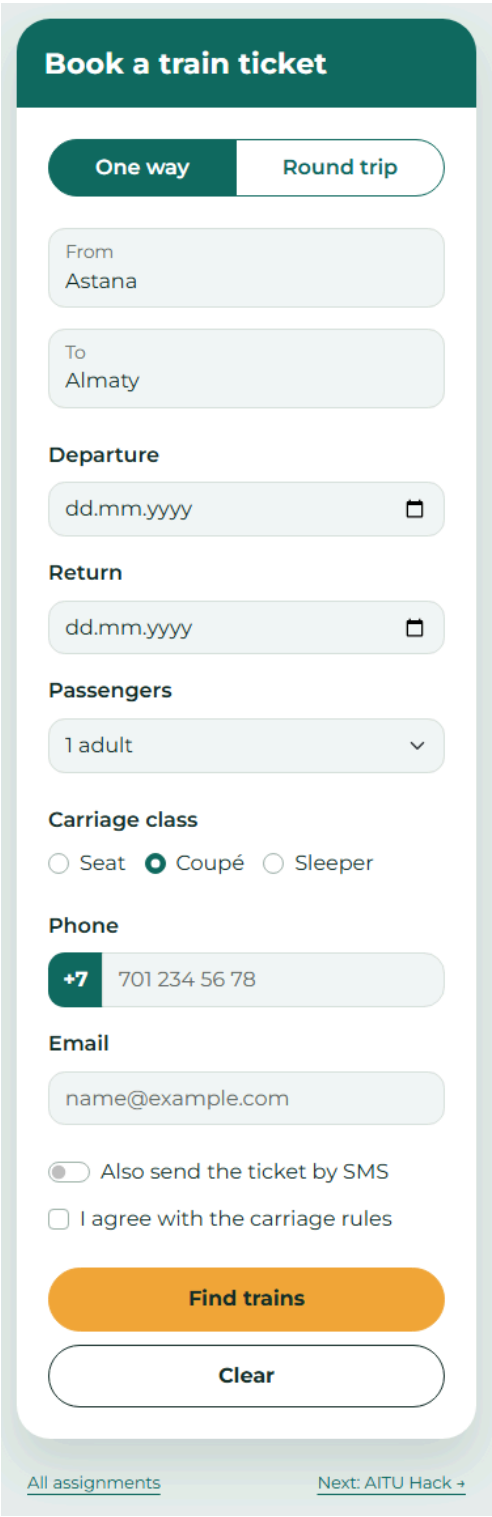
Task 1 — mobile comparison at 375 px

Columns stack vertically on a narrow screen. The flex-column-reverse flex-md-row classes place Find trains above Clear on mobile while preserving the HTML order. The reference and implementation below show the layout at 375 px.

Assignment 4.pdf reference 375 px layout



Implemented task1.html 375 px viewport



2. AITU Hack 2026 registration

The form uses appropriate input types and required, min, max, step, minlength and maxlength attributes. Validation does not depend on JavaScript; pattern is not used. Every field except GitHub is required. Email validation does not restrict the domain. The time step of 1800 seconds represents 30 minutes.

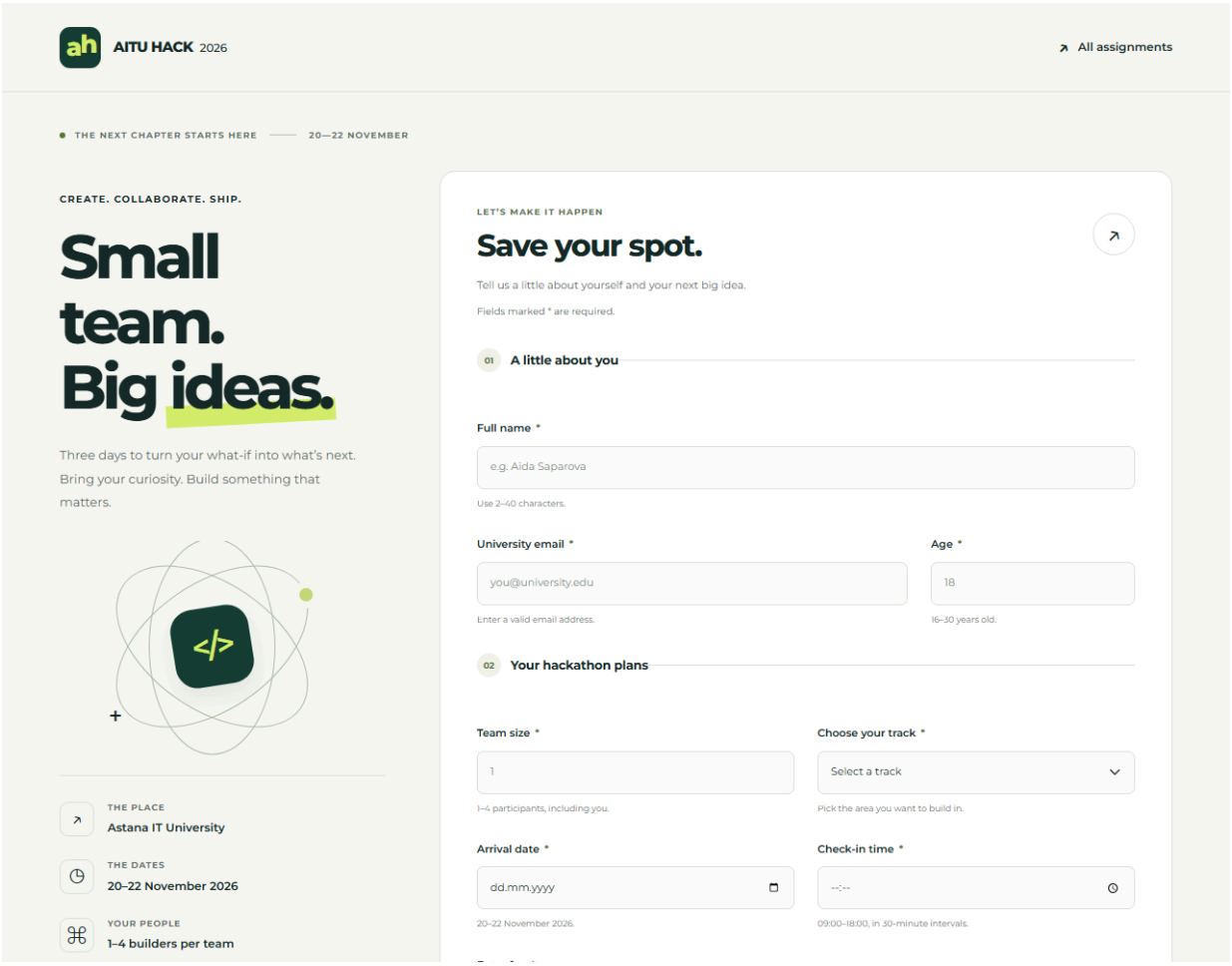


Figure 2. Task 2 in a 1280 × 1000 px viewport, showing the top of the initial form. Error messages are hidden on first load.

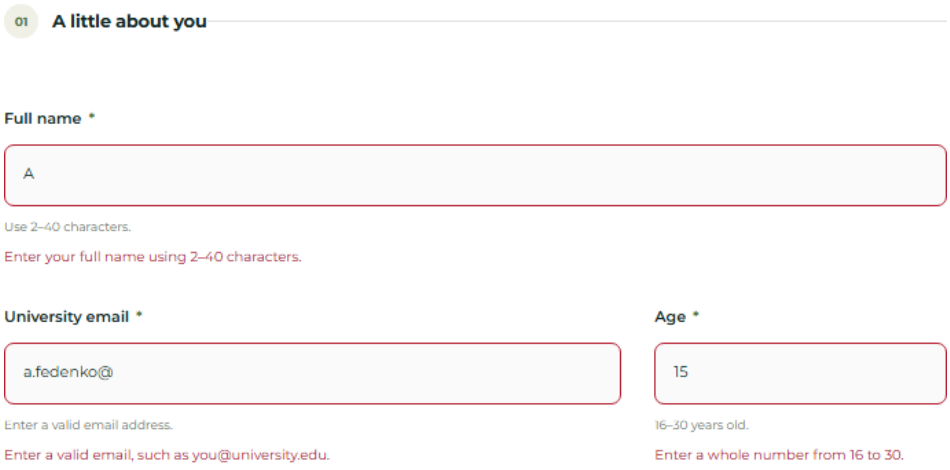


Figure 3. Name A, email a.fedenko@ and age 15 after typing and leaving the fields: a red border and invalid-feedback text explain the errors.

CSS reveals feedback using :user-invalid after interaction. Composite groups use :has(). Keyboard users receive a visible focus indicator. Hints and feedback are associated through aria-describedby.

Task 2 — assignment example results (1/2)

Browser: Chromium 145.0.7632.6. Each example was entered and the submit button was activated; all other required fields held valid values. All 32 of 32 assignment examples matched the expected result.

Field	Tested value	Expected	Actual
Full name	Anna-Maria Lee	Accept	Accepted; submit=1 (pass)
Full name	A	Reject	Rejected; submit=0 (pass)
Email	a.fedenko@astanait.edu.kz	Accept	Accepted; submit=1 (pass)
Email	a.fedenko	Reject	Rejected; submit=0 (pass)
Email	a.fedenko@	Reject	Rejected; submit=0 (pass)
Age	18	Accept	Accepted; submit=1 (pass)
Age	15	Reject	Rejected; submit=0 (pass)
Age	31	Reject	Rejected; submit=0 (pass)
Team size	4	Accept	Accepted; submit=1 (pass)
Team size	5	Reject	Rejected; submit=0 (pass)
Team size	2.5	Reject	Rejected; submit=0 (pass)
Track	Web	Accept	Accepted; submit=1 (pass)
Track	Empty option: Choose…	Reject	Rejected; submit=0 (pass)
Arrival	2026-11-21	Accept	Accepted; submit=1 (pass)
Arrival	2026-11-25	Reject	Rejected; submit=0 (pass)
Check-in	10:30	Accept	Accepted; submit=1 (pass)
Check-in	10:15	Reject	Rejected; submit=0 (pass)
Check-in	19:00	Reject	Rejected; submit=0 (pass)

submit=0 means the browser blocked submission and no submit event occurred; submit=1 means a valid form emitted the event. A test handler cancelled the actual submission during these checks. That handler is not present in the submitted HTML.

How the constraints work

minlength/maxlength enforce name length 2–40, password length 8–20 and motivation length 50–300. min/max restrict age to 16–30, team size to 1–4 and fee to 0–5000. step=1 requires integers; the fee uses step=500. Arrival is limited to 20–22 November 2026; check-in is 09:00–18:00. The initial select option has value="". Experience radios share one name. The agreement checkbox uses required.

Task 2 — assignment example results (2/2)

Field	Tested value	Expected	Actual
Entry fee	1500	Accept	Accepted; submit=1 (pass)
Entry fee	1250	Reject	Rejected; submit=0 (pass)
Entry fee	6000	Reject	Rejected; submit=0 (pass)
Experience	Beginner	Accept	Accepted; submit=1 (pass)
Experience	No selection	Reject	Rejected; submit=0 (pass)
GitHub	Empty field	Accept	Accepted; submit=1 (pass)
GitHub	https://github.com/yerassyl-dev	Accept	Accepted; submit=1 (pass)
GitHub	github.com/yerassyl-dev	Reject	Rejected; submit=0 (pass)
Password	Astana2026	Accept	Accepted; submit=1 (pass)
Password	short	Reject	Rejected; submit=0 (pass)
Motivation	120 characters	Accept	Accepted; submit=1 (pass)
Motivation	49 characters	Reject	Rejected; submit=0 (pass)
Agreement	Checked	Accept	Accepted; submit=1 (pass)
Agreement	Unchecked	Reject	Rejected; submit=0 (pass)

Boundary values and interaction

Another 23 boundary and unseen-value checks were run; 23 passed. Coverage includes range endpoints (16/30, 1, 20/22 November, 09:00/18:00 and 0/5000), values outside the ranges, fractional age, text lengths 2/40, 8/20 and 50/300, and all four tracks.

maxlength typing checks: 3/3 passed. Attempting to type an extra character leaves the length at 40, 20 or 300. These checks use keyboard input, rather than only assigning a value through a programming interface.

The browser suite passed 88/88 checks. It covers HTML attributes, assignment examples, boundaries, hidden initial errors, feedback after interaction, absence of script/novalidate, and no horizontal overflow at 375 and 1280 px. A separate keyboard and semantics suite passed 44/44 checks across the three forms.

Client-side validation checks format and stated constraints; it cannot prove that an email address or GitHub page exists. This demonstration creates no actual registration, account or payment.

3. Corrected cafe reservation form

task3.html preserves every original field: name, email, phone, guests, date, Inside / Terrace, Lunch / Dinner, comments, agreement and Reserve. The correction adds persistent associated labels, submission names, unique identifiers, appropriate input types, fieldset/legend groups and a native submit button.

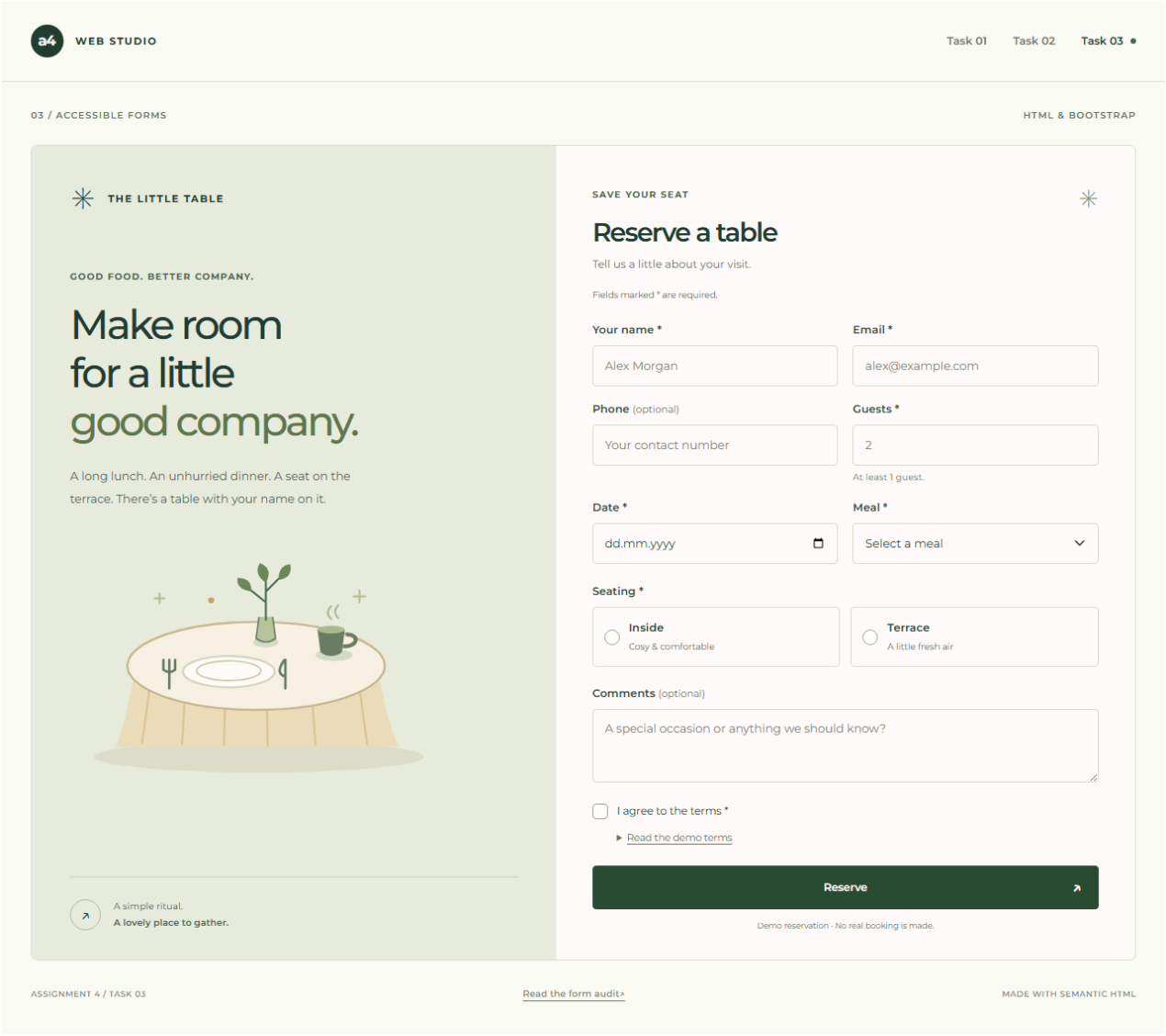


Figure 4. Corrected cafe reservation form at a viewport width of 1280 px.

Phone and comments are optional. Guests must be a positive integer. No upper capacity or date limits are imposed because the assignment does not specify them. Keyboard operation includes Tab navigation, arrow keys for radio options, Space for the checkbox and native button activation.

Task 3 — original code audit (1/2)

The table identifies 17 markup, accessibility and behavior problems. Line numbers refer to the 21-line original snippet reproduced later in this report. Each row states the user impact and implemented correction; an HTML validator alone cannot detect every issue.

No.	Line / exact fragment	User impact	Implemented correction
1	3: <input type="text" placeholder="Your name">	The name field has no persistent label; its placeholder disappears while typing.	Added id="full-name" and a visible, associated label.
2	4: <input type="text" placeholder="Email">	The email field loses its only visual identification when a value is entered.	Added id="email" and label for="email".
3	4: type="text" for Email	Arbitrary text is accepted instead of an email address; a suitable mobile keyboard is not requested.	Used type="email", autocomplete="email" and required.
4	3, 4, 6, 8, 10, 14, 18, 19: missing name; e.g. <input type="text" id="phone">	Unnamed controls do not contribute their values to a normal form submission. An id does not replace name.	Added meaningful names: full_name, email, phone, guests, date, meal, comments and terms.
5	5: <label>Phone</label>	Clicking the label does not focus the field; there is no programmatic association.	Added for="phone" matching the control id.
6	6: <input type="text" id="phone">	The browser cannot identify the telephone purpose and may display a general text keyboard.	Used type="tel" and autocomplete="tel". The tel type alone does not validate a telephone format.
7	7–8: for="guests", but id="people"	The Guests label is not associated with the guest count field.	Changed the control to id="guests" so for and id match.
8	8: type="text" for Guests	Letters, zero, negative values and fractional guest counts can be entered.	Used type="number" min="1" step="1" required.
9	10: type="text" id="date" placeholder="dd/mm/yyyy"	A placeholder cannot validate a real date; arbitrary text is accepted.	Used type="date" name="date" required for native date entry.

Task 3 — original code audit (2/2)

No.	Line / exact fragment	User impact	Implemented correction
10	10 and 12: duplicate id="date"	Duplicate identifiers make element associations ambiguous and violate id uniqueness.	Kept id="date" for the date; assigned seating-inside and seating-terrace to the radio controls.
11	12–13: name="inside" and name="terrace"	Both options can be selected at once because they belong to different radio groups.	Used a shared name="seating" with distinct value="inside" and value="terrace".
12	12–13: plain text > Inside and > Terrace	Clicking the text does not select an option; the radio controls lack associated labels.	Wrapped each option in a label, also associated through for and a unique id.
13	11–13: Seating: before the radios	A paragraph does not establish a semantic group for assistive technology.	Grouped the options in a fieldset with the legend Seating.
14	14: <select>	The Lunch / Dinner list has no associated label explaining its purpose.	Added id="meal", name="meal", a label and an empty initial option.
15	18: <textarea placeholder="Comments"> </textarea>	The comment prompt disappears while typing; there is no persistent label.	Added id, name and an associated label. Marked the comment as optional.
16	19: <input type="checkbox"> I agree to the terms	The text does not toggle the checkbox, and the control lacks an associated accessible name.	Added id="terms", name, value, label for="terms" and required. Terms are available in a details element.
17	20: Reserve	The element looks like a button but cannot submit the form and lacks native keyboard behavior.	Replaced it with <button type="submit"> for keyboard activation, validation and submission.

Missing doctype, head and body are not counted as original errors because the assignment presents a snippet. The final standalone page includes these elements, lang="en", UTF-8 and a viewport declaration. Missing form action/method is also valid HTML: the defaults use the current URL and GET.

Task 3 — alternatives and original snippet

Two alternative corrections

1. Guests, lines 7–8: retaining id="people" and changing the label to for="people" is a valid minimal fix. The implementation uses id="guests" to keep for, id and name consistent. The isolated assignment snippet has no dependencies on the old id; an existing project would need those dependencies checked before renaming it.

2. Reserve, line 20: <input type="submit" value="Reserve"> would also provide native submission. A button type="submit" was chosen because it can contain both text and a decorative aria-hidden arrow while retaining an accessible name. Adding only role="button" and tabindex to the original div would not provide submission or keyboard handling.

Original 21-line snippet

```
<h2>Reserve a table</h2>
<form>
  <input type="text" placeholder="Your name">
  <input type="text" placeholder="Email">
  <label>Phone</label>
  <input type="text" id="phone">
  <label for="guests">Guests</label>
  <input type="text" id="people">
  <label for="date">Date</label>
  <input type="text" id="date" placeholder="dd/mm/yyyy">
  <p>Seating:</p>
  <input type="radio" name="inside" id="date"> Inside
  <input type="radio" name="terrace"> Terrace
  <select>
    <option>Lunch</option>
    <option>Dinner</option>
  </select>
  <textarea placeholder="Comments"></textarea>
  <input type="checkbox"> I agree to the terms
  <div class="btn btn-primary">Reserve</div>
</form>
```

Validation and conclusion

W3C Nu HTML Checker

Page	Actual result	Check time
task1.html	Errors: 0; warnings: 0. Passed.	2026-10-17 13:05:45 UTC
task2.html	Errors: 0; warnings: 0. Passed.	2026-10-17 13:04:45 UTC
task3.html	Errors: 0; warnings: 0. Passed.	2026-10-17 13:19:05 UTC
index.html	Errors: 0; warnings: 0. Passed.	2026-10-17 13:19:05 UTC
completed.html	Errors: 0; warnings: 0. Passed.	2026-10-17 13:04:45 UTC

These results come from saved responses of the official W3C Nu HTML Checker at https://validator.w3.org/nu/. HTML validation complements the keyboard and form-behavior checks; it does not replace them.

Running the demonstration

Run python serve.py, then open http://127.0.0.1:8000/. The forms POST to completed.html. The local server reads and discards the submitted data, then redirects to the completion page. It does not store data, issue tickets or reserve tables. A production service would also need server-side validation and business logic.

Conclusion

The three tasks demonstrate responsive Bootstrap layouts, native HTML validation and feedback after user interaction. Associated labels, shared radio groups, meaningful names and native buttons improve accessibility and submission behavior. Boundary testing complements the assignment examples and verifies behavior with new values.

Sources

Assignment brief and reference images: Assignment 4.pdf, supplied for this coursework. Technical documentation:

Bootstrap 5.3: forms and components

Bootstrap 5.3: grid

MDN: HTML constraint validation

MDN: input type=time and step in seconds

MDN: :user-invalid

MDN: associating a label with a form control

W3C Nu HTML Checker